

MTH253 – Calculus III Notes

Chapters 1 & 2: Multivariable Differentiation & Multiple Integration

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1 Chapter 1: Multivariable Differentiation

1.1 Coordinates and Surfaces in $\mathbb{R}^3$

Key Notations

- $\mathbb{R}^3$: Three-dimensional real coordinate space, representing points (x, y, z) .
- $d(P_1, P_2)$: The Euclidean distance between points P_1 and P_2 .
- S : A surface represented by a single equation in x, y, z .

Concise Explanation & Formulas

In 3D space, an equation with one constraint generally defines a surface.

Key Formula(s)

Distance Formula in $\mathbb{R}^3$:

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Equation of a Sphere: A sphere with center $C(h, k, l)$ and radius R :

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = R^2$$

Cylindrical Surfaces: Surfaces whose equations contain only two variables (e.g., $x^2 + y^2 = R^2$) project as a curve in 2D and extend infinitely parallel to the axis of the missing variable (z -axis here) in $\mathbb{R}^3$.

Example 1.1: Distance and Cylindrical Surfaces

Part A: Distance and Sphere

Find the equation of the sphere centered at $C(2, -1, 3)$ that passes through $P(4, 2, -1)$.

Solution:

First, find the radius R as the distance from C to P :

$$R = d(C, P) = \sqrt{(4 - 2)^2 + (2 - (-1))^2 + (-1 - 3)^2} = \sqrt{2^2 + 3^2 + (-4)^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$$

The equation of the sphere is:

$$(x - 2)^2 + (y + 1)^2 + (z - 3)^2 = 29$$

Part B: Cylindrical Surface

Describe the surface represented by $x^2 + y^2 = 1$ in $\mathbb{R}^3$.

Solution:

The equation $x^2 + y^2 = 1$ does not restrict z . In the xy -plane, this is a circle of radius 1 centered at the origin. Since z is free, we replicate this circle at all values of z , which forms a circular cylinder of radius 1 whose central axis is the z -axis.

1.2 Functions of Several Variables

Key Notations

- $f : D \rightarrow \mathbb{R}$: A function mapping a domain $D \subset \mathbb{R}^n$ to $\mathbb{R}$.
- $D(f)$: The domain of f , the set of all inputs for which f is well-defined.
- Level Curves: Contours in 2D where $f(x, y) = k$ (for constant k).
- Level Surfaces: Surfaces in 3D where $f(x, y, z) = k$.

Concise Explanation & Formulas

A multivariable function maps a tuple of variables to a single real output. The graph of $z = f(x, y)$ is a surface in $\mathbb{R}^3$.

Key Formula(s)

Level Curves: The set of points in the domain of $f(x, y)$ satisfying:

$$\{(x, y) \in D \mid f(x, y) = k\}, \quad k \in \text{Range}(f)$$

Level Surfaces: The set of points in the domain of $f(x, y, z)$ satisfying:

$$\{(x, y, z) \in D \mid f(x, y, z) = k\}$$

Example 1.2: Domain and Level Curves

Let $f(x, y) = \sqrt{9 - x^2 - y^2}$.

1. Find the domain and range of f .
2. Sketch the level curves for $k = 0, \sqrt{5}, 3$.

Solution:

1. For the square root to be defined, we must have $9 - x^2 - y^2 \geq 0 \implies x^2 + y^2 \leq 9$. Thus, the domain D is the closed disk centered at the origin with radius 3:

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 9\}$$

The range is $[0, 3]$ because $0 \leq \sqrt{9 - x^2 - y^2} \leq \sqrt{9} = 3$.

2. The level curves are given by $\sqrt{9 - x^2 - y^2} = k \implies x^2 + y^2 = 9 - k^2$.

- For $k = 0$: $x^2 + y^2 = 9$ (circle of radius 3).
- For $k = \sqrt{5}$: $x^2 + y^2 = 9 - 5 = 4$ (circle of radius 2).
- For $k = 3$: $x^2 + y^2 = 9 - 9 = 0$ (the origin $(0, 0)$).

1.3 Limits and Continuity

Key Notations

- $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$: The limit of f as (x, y) approaches (a, b) is L .

- C_1, C_2 : Paths of approach in the input plane.

Concise Explanation & Formulas

For a limit to exist in $\mathbb{R}^2$, the function must approach the same value L along *every possible path* approaching (a, b) . If we find two paths yielding different limits, the limit does not exist.

Key Formula(s)

Limit Non-Existence Test: If $f(x, y) \rightarrow L_1$ along path C_1 and $f(x, y) \rightarrow L_2$ along path C_2 as $(x, y) \rightarrow (a, b)$, and $L_1 \neq L_2$, then:

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) \text{ does not exist.}$$

Squeeze Theorem

If $g(x, y) \leq f(x, y) \leq h(x, y)$ for all (x, y) near (a, b) and $\lim_{(x,y) \rightarrow (a,b)} g(x, y) = \lim_{(x,y) \rightarrow (a,b)} h(x, y) = L$, then:

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

Example 1.3: Evaluating Limits and Non-Existence

Part A: Non-existence

Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Solution:

Test path C_1 along the x -axis ($y = 0$):

$$\lim_{(x,0) \rightarrow (0,0)} \frac{x^2 - 0^2}{x^2 + 0^2} = \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1$$

Test path C_2 along the y -axis ($x = 0$):

$$\lim_{(0,y) \rightarrow (0,0)} \frac{0^2 - y^2}{0^2 + y^2} = \lim_{y \rightarrow 0} \frac{-y^2}{y^2} = -1$$

Since the two limits are different ($1 \neq -1$), the limit does not exist.

Part B: Squeeze Theorem

Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2}$.

Solution:

Notice that $x^2 \leq x^2 + y^2 \implies \frac{x^2}{x^2 + y^2} \leq 1$. Taking absolute values:

$$0 \leq \left| \frac{3x^2y}{x^2 + y^2} \right| = 3|y| \frac{x^2}{x^2 + y^2} \leq 3|y|$$

Therefore:

$$-3|y| \leq \frac{3x^2y}{x^2 + y^2} \leq 3|y|$$

Since $\lim_{(x,y) \rightarrow (0,0)} (-3|y|) = 0$ and $\lim_{(x,y) \rightarrow (0,0)} 3|y| = 0$, by the Squeeze Theorem:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2} = 0$$

1.4 Partial Derivatives

Key Notations

- $f_x = \frac{\partial f}{\partial x}$: Partial derivative of f with respect to x (holding y constant).
- $f_y = \frac{\partial f}{\partial y}$: Partial derivative of f with respect to y (holding x constant).
- $f_{xx}, f_{yy}, f_{xy}, f_{yx}$: Second-order partial derivatives.

Concise Explanation & Formulas

Partial derivatives measure the rates of change of a multivariable function along the coordinate directions.

Key Formula(s)

First-Order Partial Derivatives:

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}, \quad f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}$$

Second-Order Mixed Partial Derivatives:

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right), \quad f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$$

Clairaut's Theorem

If f is defined on a disk D containing (a, b) , and mixed partial derivatives f_{xy} and f_{yx} are both continuous on D , then:

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Example 1.4: First and Second Order Partial Derivatives

Let $f(x, y) = x^3 + x^2y^3 - 2y^2$. Find all first and second order partial derivatives, and verify Clairaut's Theorem.

Solution:

First-order partial derivatives:

$$f_x = \frac{\partial}{\partial x}(x^3 + x^2y^3 - 2y^2) = 3x^2 + 2xy^3$$

$$f_y = \frac{\partial}{\partial y}(x^3 + x^2y^3 - 2y^2) = 3x^2y^2 - 4y$$

Second-order partial derivatives:

$$f_{xx} = \frac{\partial}{\partial x}(3x^2 + 2xy^3) = 6x + 2y^3$$

$$f_{yy} = \frac{\partial}{\partial y}(3x^2y^2 - 4y) = 6x^2y - 4$$

Mixed partial derivatives:

$$f_{xy} = \frac{\partial}{\partial y}(3x^2 + 2xy^3) = 6xy^2$$

$$f_{yx} = \frac{\partial}{\partial x}(3x^2y^2 - 4y) = 6xy^2$$

Since $f_{xy} = f_{yx} = 6xy^2$ and this expression is continuous everywhere, Clairaut's Theorem is verified.

1.5 Tangent Planes and Linearization

Key Notations

- $L(x, y)$: Linearization of $f(x, y)$ at (a, b) .
- dz : The total differential of $z = f(x, y)$.

Concise Explanation & Formulas

Near a point where a function is differentiable, the surface $z = f(x, y)$ can be approximated by its tangent plane. The equation of the tangent plane defines the linear approximation (linearization) of the function.

Key Formula(s)

Tangent Plane Equation: The tangent plane to $z = f(x, y)$ at $P(x_0, y_0, z_0)$ is:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Linearization and Linear Approximation:

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \implies f(x, y) \approx L(x, y)$$

Total Differential: For $z = f(x, y)$, the differential dz is:

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy = f_x(x, y)dx + f_y(x, y)dy$$

Example 1.5: Tangent Plane and Linearization

For $f(x, y) = 2x^2 + y^2$, find the tangent plane and the linearization at $(1, 1)$. Then approximate $f(1.05, 0.95)$.

Solution:

First, find the function value and partial derivatives at $(1, 1)$:

$$f(1, 1) = 2(1)^2 + (1)^2 = 3 \implies z_0 = 3$$

$$f_x = 4x \implies f_x(1, 1) = 4$$

$$f_y = 2y \implies f_y(1, 1) = 2$$

The tangent plane equation is:

$$z - 3 = 4(x - 1) + 2(y - 1) \implies z = 4x + 2y - 3$$

The linearization is:

$$L(x, y) = 3 + 4(x - 1) + 2(y - 1)$$

We approximate $f(1.05, 0.95)$ using $L(1.05, 0.95)$:

$$f(1.05, 0.95) \approx 3 + 4(1.05 - 1) + 2(0.95 - 1) = 3 + 4(0.05) + 2(-0.05) = 3 + 0.20 - 0.10 = 3.10$$

(The exact value is $2(1.05)^2 + (0.95)^2 = 2.205 + 0.9025 = 3.1075$, showing the approximation is very close.)

1.6 The Chain Rule and Implicit Differentiation

Key Notations

- Tree Diagram: Used to trace dependency of variables.
- $\frac{\partial z}{\partial s}$, $\frac{\partial z}{\partial t}$: Partial derivatives when variables depend on multiple parameters.

Concise Explanation & Formulas

When variables themselves are functions of other variables, we use the chain rule. We sum the products of derivatives along all paths in the dependency tree.

Key Formula(s)

Chain Rule: Case 1 (One Parameter): If $z = f(x, y)$ and $x = x(t)$, $y = y(t)$, then:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Chain Rule: Case 2 (Two Parameters): If $z = f(x, y)$ and $x = x(s, t)$, $y = y(s, t)$, then:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}, \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

Implicit Differentiation: If $F(x, y, z) = 0$ defines $z = f(x, y)$ implicitly, then:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} \quad (\text{for } F_z \neq 0)$$

Example 1.6: Chain Rule and Implicit Differentiation

Part A: Chain Rule

If $z = x^2y + 3xy^4$, where $x = \sin(2t)$ and $y = \cos(t)$, find $\frac{dz}{dt}$ at $t = 0$.

Solution:

At $t = 0$: $x = \sin(0) = 0$ and $y = \cos(0) = 1$.

Find the partial derivatives of z :

$$\frac{\partial z}{\partial x} = 2xy + 3y^4 \implies \left. \frac{\partial z}{\partial x} \right|_{(0,1)} = 2(0)(1) + 3(1)^4 = 3$$

$$\frac{\partial z}{\partial y} = x^2 + 12xy^3 \implies \left. \frac{\partial z}{\partial y} \right|_{(0,1)} = 0^2 + 12(0)(1)^3 = 0$$

Find the derivatives of x and y at $t = 0$:

$$\frac{dx}{dt} = 2\cos(2t) \implies \left. \frac{dx}{dt} \right|_{t=0} = 2, \quad \frac{dy}{dt} = -\sin(t) \implies \left. \frac{dy}{dt} \right|_{t=0} = 0$$

Apply Case 1 of the Chain Rule:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = (3)(2) + (0)(0) = 6$$

Part B: Implicit Differentiation

Find $\frac{\partial z}{\partial x}$ if $x^3 + y^3 + z^3 + 6xyz = 1$.

Solution:

Define $F(x, y, z) = x^3 + y^3 + z^3 + 6xyz - 1 = 0$.

Find the partial derivatives of F :

$$F_x = 3x^2 + 6yz, \quad F_z = 3z^2 + 6xy$$

Use the implicit differentiation formula:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{3x^2 + 6yz}{3z^2 + 6xy} = -\frac{x^2 + 2yz}{z^2 + 2xy}$$

1.7 Gradients and Directional Derivatives

Key Notations

- ∇f : The gradient vector of f .
- $D_{\vec{u}}f$: The directional derivative of f in the direction of the unit vector $\vec{u}$.

Concise Explanation & Formulas

The gradient ∇f points in the direction of maximum rate of increase of the function. The directional derivative calculates the rate of change of f along any arbitrary direction vector $\vec{u}$.

Key Formula(s)

Gradient Vector:

$$\nabla f(x, y) = \langle f_x, f_y \rangle = f_x \mathbf{i} + f_y \mathbf{j}, \quad \nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle$$

Directional Derivative (using unit vector $\vec{u}$):

$$D_{\vec{u}}f(x, y) = \nabla f(x, y) \cdot \vec{u}$$

If given a general direction vector $\vec{v}$, you **must** normalize it first: $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$.

Properties of the Gradient:

- The maximum value of $D_{\vec{u}}f(x, y)$ is $\|\nabla f(x, y)\|$, occurring when $\vec{u}$ is in the direction of ∇f .
- The minimum value is $-\|\nabla f(x, y)\|$, in the direction of $-\nabla f$.
- The gradient vector $\nabla f(x_0, y_0)$ is always orthogonal to the level curve $f(x, y) = k$ passing through (x_0, y_0) .

Example 1.7: Directional Derivative and Maximum Rate of Change

Let $f(x, y) = x^2y^3 - 4y$.

1. Find the directional derivative of f at $P(2, -1)$ in the direction of $\vec{v} = \langle 2, 5 \rangle$.
2. Find the maximum rate of change of f at $P(2, -1)$ and the direction in which it occurs.

Solution:

1. Calculate the gradient:

$$\nabla f(x, y) = \langle 2xy^3, 3x^2y^2 - 4 \rangle$$

At $P(2, -1)$:

$$\nabla f(2, -1) = \langle 2(2)(-1)^3, 3(2)^2(-1)^2 - 4 \rangle = \langle -4, 12 - 4 \rangle = \langle -4, 8 \rangle$$

Normalize the direction vector $\vec{v}$:

$$\|\vec{v}\| = \sqrt{2^2 + 5^2} = \sqrt{29} \implies \vec{u} = \left\langle \frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right\rangle$$

The directional derivative is:

$$D_{\vec{u}}f(2, -1) = \nabla f(2, -1) \cdot \vec{u} = -4 \left(\frac{2}{\sqrt{29}} \right) + 8 \left(\frac{5}{\sqrt{29}} \right) = \frac{-8 + 40}{\sqrt{29}} = \frac{32}{\sqrt{29}}$$

2. The maximum rate of change is:

$$\|\nabla f(2, -1)\| = \|\langle -4, 8 \rangle\| = \sqrt{(-4)^2 + 8^2} = \sqrt{16 + 64} = \sqrt{80} = 4\sqrt{5}$$

This maximum rate of change occurs in the direction of the gradient vector $\nabla f(2, -1) = \langle -4, 8 \rangle$.

1.8 Extrema of Multivariable Functions

Key Notations

- Critical Point: A point (a, b) where $f_x(a, b) = 0$ and $f_y(a, b) = 0$ (or either is undefined).
- D : The discriminant (Hessian determinant) used in the Second Derivatives Test.
- Saddle Point: A critical point that is neither a local max nor local min.

Concise Explanation & Formulas

To locate local extrema, we find critical points and classify them using second-order partial derivatives.

Key Formula(s)

Second Derivatives Test: Let (a, b) be a critical point of f where second partials are continuous. Let:

$$D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

- If $D > 0$ and $f_{xx}(a, b) > 0 \implies f(a, b)$ is a **local minimum**.
- If $D > 0$ and $f_{xx}(a, b) < 0 \implies f(a, b)$ is a **local maximum**.
- If $D < 0 \implies (a, b)$ is a **saddle point**.
- If $D = 0 \implies$ the test is **inconclusive** (requires alternative analysis).

Example 1.8: Finding Local Extrema and Saddle Points

Find the local maximum and minimum values and saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$.

Solution:

Step 1: Find critical points.

$$f_x = 4x^3 - 4y = 0 \implies y = x^3$$

$$f_y = 4y^3 - 4x = 0 \implies x = y^3$$

Substitute $y = x^3$ into the second equation:

$$x = (x^3)^3 = x^9 \implies x(x^8 - 1) = 0$$

This gives real solutions $x = -1, 0, 1$.

- If $x = 0, y = 0 \implies (0, 0)$.
- If $x = 1, y = 1 \implies (1, 1)$.
- If $x = -1, y = -1 \implies (-1, -1)$.

Step 2: Apply the Second Derivatives Test. Calculate second partials:

$$f_{xx} = 12x^2, \quad f_{yy} = 12y^2, \quad f_{xy} = -4$$

The discriminant is $D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = (12x^2)(12y^2) - (-4)^2 = 144x^2y^2 - 16$.

- At $(0, 0)$: $D(0, 0) = -16 < 0$. Thus, $(0, 0)$ is a **saddle point**.
- At $(1, 1)$: $D(1, 1) = 144(1)(1) - 16 = 128 > 0$ and $f_{xx}(1, 1) = 12 > 0$. Thus, $f(1, 1) = -1$ is a **local minimum**.
- At $(-1, -1)$: $D(-1, -1) = 128 > 0$ and $f_{xx}(-1, -1) = 12 > 0$. Thus, $f(-1, -1) = -1$ is a **local minimum**.

1.9 Lagrange Multipliers

Key Notations

- $f(x, y, z)$: The objective function to maximize or minimize.
- $g(x, y, z) = k$: The constraint equation.
- λ : The Lagrange multiplier (scalar).

Concise Explanation & Formulas

Lagrange multipliers find the extrema of an objective function subject to constraints. Geometrically, at an extreme point, the level curves of f and g are tangent, meaning their normal vectors (gradients) are parallel.

Key Formula(s)

Lagrange System of Equations (1 Constraint): Solve for x, y, z and λ :

$$\begin{cases} \nabla f = \lambda \nabla g \\ g(x, y, z) = k \end{cases} \iff \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ f_z = \lambda g_z \\ g(x, y, z) = k \end{cases}$$

Example 1.9: Extrema with Lagrange Multipliers

Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle $x^2 + y^2 = 1$.

Solution:

Let the constraint be $g(x, y) = x^2 + y^2 = 1$. The gradient vectors are:

$$\nabla f = \langle 2x, 4y \rangle, \quad \nabla g = \langle 2x, 2y \rangle$$

Write down the system of equations:

$$2x = \lambda(2x) \implies x(1 - \lambda) = 0 \tag{1}$$

$$4y = \lambda(2y) \implies y(2 - \lambda) = 0 \tag{2}$$

$$x^2 + y^2 = 1 \tag{3}$$

From equation (1), either $x = 0$ or $\lambda = 1$.

- Case A: $x = 0$. Substituting into (3) yields $y^2 = 1 \implies y = \pm 1$. This gives points $(0, 1)$ and $(0, -1)$.
- Case B: $\lambda = 1$. Substituting $\lambda = 1$ into (2) yields $y(2 - 1) = 0 \implies y = 0$. Substituting into (3) yields $x^2 = 1 \implies x = \pm 1$. This gives points $(1, 0)$ and $(-1, 0)$.

Evaluate $f(x, y)$ at these candidate points:

- $f(0, \pm 1) = 0^2 + 2(\pm 1)^2 = 2$ (Absolute Maximum).
- $f(\pm 1, 0) = (\pm 1)^2 + 2(0)^2 = 1$ (Absolute Minimum).

2 Chapter 2: Multiple Integration

2.1 Double Integrals over Rectangles

Key Notations

- $\iint_R f(x, y) dA$: Double integral of f over rectangle R .
- $dA = dy dx$ or $dx dy$: Area element.

Concise Explanation & Formulas

A double integral over a rectangle sums function values weighted by sub-rectangle areas. Fubini's Theorem states that we can evaluate a double integral over a rectangle as an iterated single integral in either order.

Fubini's Theorem for Rectangles

If $f(x, y)$ is continuous on the rectangular region $R = [a, b] \times [c, d]$, then:

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

If the integrand is separable, i.e., $f(x, y) = g(x)h(y)$, then:

$$\iint_R g(x)h(y) dA = \left(\int_a^b g(x) dx \right) \left(\int_c^d h(y) dy \right)$$

Example 2.1: Evaluating Iterated Integrals

Evaluate $\iint_R (x - 3y^2) dA$ where $R = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 2, 1 \leq y \leq 2\}$.

Solution:

Using Fubini's Theorem, integrate first with respect to y :

$$\begin{aligned} \iint_R (x - 3y^2) dA &= \int_0^2 \int_1^2 (x - 3y^2) dy dx = \int_0^2 [xy - y^3]_{y=1}^{y=2} dx \\ &= \int_0^2 ((2x - 8) - (x - 1)) dx = \int_0^2 (x - 7) dx \\ &= \left[\frac{1}{2}x^2 - 7x \right]_0^2 = \left(\frac{1}{2}(4) - 14 \right) - 0 = 2 - 14 = -12 \end{aligned}$$

2.2 Double Integrals over General Regions

Key Notations

- Type I Region: Bounded above and below by continuous functions of x .
- Type II Region: Bounded left and right by continuous functions of y .

Concise Explanation & Formulas

For regions that are not simple rectangles, we set the variable bounds of the inner integral to be functions of the outer variable.

Key Formula(s)

Type I Region: $D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

Type II Region: $D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$

$$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

Area of General Region:

$$A(D) = \iint_D 1 dA$$

Example 2.2: Double Integral over General Region

Evaluate $\iint_D (x + 2y) dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

Solution:

Step 1: Find the intersection points of the boundaries to get the bounds for x .

$$2x^2 = 1 + x^2 \implies x^2 = 1 \implies x = \pm 1$$

For $x \in [-1, 1]$, we have $1 + x^2 \geq 2x^2$. Thus, D is a Type I region described by:

$$D = \{(x, y) \mid -1 \leq x \leq 1, 2x^2 \leq y \leq 1 + x^2\}$$

Step 2: Set up and evaluate the integral.

$$\begin{aligned} \iint_D (x + 2y) dA &= \int_{-1}^1 \int_{2x^2}^{1+x^2} (x + 2y) dy dx = \int_{-1}^1 [xy + y^2]_{y=2x^2}^{y=1+x^2} dx \\ &= \int_{-1}^1 (x(1+x^2) + (1+x^2)^2 - x(2x^2) - (2x^2)^2) dx \\ &= \int_{-1}^1 (x + x^3 + 1 + 2x^2 + x^4 - 2x^3 - 4x^4) dx \\ &= \int_{-1}^1 (-3x^4 - x^3 + 2x^2 + x + 1) dx \end{aligned}$$

By symmetry, integrals of odd functions (x^3, x) over symmetric intervals $[-1, 1]$ are zero. Hence:

$$\int_{-1}^1 (-3x^4 - x^3 + 2x^2 + x + 1) dx = 2 \int_0^1 (-3x^4 + 2x^2 + 1) dx$$

$$= 2 \left[-\frac{3}{5}x^5 + \frac{2}{3}x^3 + x \right]_0^1 = 2 \left(-\frac{3}{5} + \frac{2}{3} + 1 \right) = 2 \left(\frac{-9 + 10 + 15}{15} \right) = \frac{32}{15}$$

2.3 Double Integrals in Polar Coordinates

Key Notations

- (r, θ) : Polar coordinates.
- $dA = r dr d\theta$: The area element in polar coordinates. (Do not forget the extra factor of r !)

Concise Explanation & Formulas

Polar coordinates are useful when boundaries of a region are circular or are parameterized naturally by distance from the origin r and angle θ .

Key Formula(s)

Coordinate Transformations:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2$$

Polar Double Integral (General Region): If $D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$, then:

$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta$$

Example 2.3: Double Integral in Polar Coordinates

Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 2x$.

Solution:

The cylinder is $x^2 + y^2 = 2x$. In polar coordinates, this is:

$$r^2 = 2r \cos \theta \implies r = 2 \cos \theta$$

To lie inside the cylinder, r goes from 0 to $2 \cos \theta$. For $r \geq 0$, we require $\cos \theta \geq 0 \implies \theta \in [-\pi/2, \pi/2]$. Thus, the polar region of integration D is:

$$D = \{(r, \theta) \mid -\pi/2 \leq \theta \leq \pi/2, 0 \leq r \leq 2 \cos \theta\}$$

The volume is the integral of the height $z = x^2 + y^2 = r^2$:

$$\begin{aligned} V &= \iint_D (x^2 + y^2) dA = \int_{-\pi/2}^{\pi/2} \int_0^{2 \cos \theta} (r^2) r dr d\theta = \int_{-\pi/2}^{\pi/2} \int_0^{2 \cos \theta} r^3 dr d\theta \\ &= \int_{-\pi/2}^{\pi/2} \left[\frac{1}{4} r^4 \right]_0^{2 \cos \theta} d\theta = \int_{-\pi/2}^{\pi/2} 4 \cos^4 \theta d\theta \end{aligned}$$

Using symmetry and the half-angle identity $\cos^2 \theta = \frac{1+\cos(2\theta)}{2}$:

$$\cos^4 \theta = \left(\frac{1 + \cos(2\theta)}{2} \right)^2 = \frac{1}{4}(1 + 2\cos(2\theta) + \cos^2(2\theta)) = \frac{1}{4} \left(1 + 2\cos(2\theta) + \frac{1 + \cos(4\theta)}{2} \right)$$

Therefore:

$$4\cos^4 \theta = \frac{3}{2} + 2\cos(2\theta) + \frac{1}{2}\cos(4\theta)$$

Now integrate:

$$V = \int_{-\pi/2}^{\pi/2} \left(\frac{3}{2} + 2\cos(2\theta) + \frac{1}{2}\cos(4\theta) \right) d\theta = \left[\frac{3}{2}\theta + \sin(2\theta) + \frac{1}{8}\sin(4\theta) \right]_{-\pi/2}^{\pi/2} = \frac{3}{2}\pi$$

2.4 Triple Integrals

Key Notations

- $\iiint_E f(x, y, z) dV$: Triple integral over a 3D solid E .
- $dV = dz dy dx$: Volume element in Cartesian coordinates.
- $V(E)$: Volume of solid E .

Concise Explanation & Formulas

Triple integrals compute quantities (like mass or volume) over 3D domains. For a Type 1 solid region, we integrate along z first, between lower and upper bounding surfaces $u_1(x, y)$ and $u_2(x, y)$, and then integrate over the 2D projection D on the xy -plane.

Key Formula(s)

Fubini's Theorem for a Rectangular Box $B = [a, b] \times [c, d] \times [r, s]$:

$$\iiint_B f(x, y, z) dV = \int_r^s \int_c^d \int_a^b f(x, y, z) dx dy dz$$

Integration over Type 1 Solid Region: If $E = \{(x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$, then:

$$\iiint_E f(x, y, z) dV = \iint_D \left(\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz \right) dA$$

Volume of a Solid:

$$V(E) = \iiint_E 1 dV$$

Example 2.4: Triple Integral and Tetrahedron Volume

Evaluate $\iiint_E z \, dV$, where E is the solid tetrahedron bounded by the coordinate planes $x = 0$, $y = 0$, $z = 0$, and the plane $x + y + z = 1$.

Solution:

The upper surface is $z = 1 - x - y$ and the lower surface is $z = 0$. The projection D on the xy -plane is bounded by $x = 0$, $y = 0$, and $x + y = 1 \implies y = 1 - x$. Thus, the region E is described by:

$$E = \{(x, y, z) \mid 0 \leq x \leq 1, 0 \leq y \leq 1 - x, 0 \leq z \leq 1 - x - y\}$$

Set up and evaluate the triple integral:

$$\begin{aligned} \iiint_E z \, dV &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx = \int_0^1 \int_0^{1-x} \left[\frac{1}{2} z^2 \right]_0^{1-x-y} dy \, dx \\ &= \frac{1}{2} \int_0^1 \int_0^{1-x} (1 - x - y)^2 dy \, dx \end{aligned}$$

Let $u = 1 - x - y \implies dy = -du$. When $y = 0$, $u = 1 - x$. When $y = 1 - x$, $u = 0$.

$$\int_0^{1-x} (1 - x - y)^2 dy = \int_{1-x}^0 u^2 (-du) = \int_0^{1-x} u^2 du = \left[\frac{1}{3} u^3 \right]_0^{1-x} = \frac{1}{3} (1 - x)^3$$

Now compute the outer integral:

$$\iiint_E z \, dV = \frac{1}{2} \int_0^1 \frac{1}{3} (1 - x)^3 dx = \frac{1}{6} \left[-\frac{1}{4} (1 - x)^4 \right]_0^1 = \frac{1}{6} \left(0 - \left(-\frac{1}{4} \right) \right) = \frac{1}{24}$$

2.5 Triple Integrals in Cylindrical Coordinates

Key Notations

- (r, θ, z) : Cylindrical coordinates.
- $dV = r \, dz \, dr \, d\theta$: The volume element. (Includes the scaling factor r).

Concise Explanation & Formulas

Cylindrical coordinates combine polar coordinates in the xy -plane with standard linear height z . They are ideal for solids with axial symmetry around the z -axis.

Key Formula(s)

Coordinate Transformations:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z, \quad x^2 + y^2 = r^2$$

Cylindrical Integration Formula:

$$\iiint_E f(x, y, z) dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) r dz dr d\theta$$

Example 2.5: Triple Integral in Cylindrical Coordinates

Evaluate $\iiint_E \sqrt{x^2 + y^2} dV$, where E is the region that lies within the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$.

Solution:

In cylindrical coordinates, the boundaries are $r = 1$ (cylinder), $z = 4$ (top plane), and $z = 1 - r^2$ (paraboloid). The solid region is $E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 1, 1 - r^2 \leq z \leq 4\}$. The integrand is $\sqrt{x^2 + y^2} = r$. Set up the integral:

$$\begin{aligned} \iiint_E \sqrt{x^2 + y^2} dV &= \int_0^{2\pi} \int_0^1 \int_{1-r^2}^4 (r) r dz dr d\theta = \int_0^{2\pi} \int_0^1 r^2 (4 - (1 - r^2)) dr d\theta \\ &= 2\pi \int_0^1 r^2 (3 + r^2) dr = 2\pi \int_0^1 (3r^2 + r^4) dr \\ &= 2\pi \left[r^3 + \frac{1}{5} r^5 \right]_0^1 = 2\pi \left(1 + \frac{1}{5} \right) = \frac{12\pi}{5} \end{aligned}$$

2.6 Triple Integrals in Spherical Coordinates**Key Notations**

- (ρ, θ, ϕ) : Spherical coordinates.
- ρ : Distance from origin ($\rho \geq 0$).
- θ : Angle in the xy -plane ($0 \leq \theta \leq 2\pi$).
- ϕ : Angle from the positive z -axis ($0 \leq \phi \leq \pi$).
- $dV = \rho^2 \sin \phi d\rho d\theta d\phi$: The volume element. (Includes scaling factor $\rho^2 \sin \phi$).

Concise Explanation & Formulas

Spherical coordinates define a point by distance ρ and two angles θ and ϕ . They are highly efficient for spheres, cones, or regions symmetrical about the origin.

Key Formula(s)**Coordinate Transformations:**

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi, \quad x^2 + y^2 + z^2 = \rho^2$$

Spherical Integration Formula (over a wedge): If $E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$, then:

$$\iiint_E f(x, y, z) dV = \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin \phi \cos \theta, \dots) \rho^2 \sin \phi d\rho d\theta d\phi$$

Example 2.6: Volume using Spherical Coordinates

Find the volume of the solid E that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$.

Solution:

Step 1: Write the boundaries in spherical coordinates.

- Sphere: $\rho^2 = \rho \cos \phi \implies \rho = \cos \phi$.
- Cone: $\rho \cos \phi = \sqrt{\rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta)} = \rho \sin \phi \implies \tan \phi = 1 \implies \phi = \pi/4$.

Since the solid lies above the cone, the angle ϕ ranges from 0 (the positive z -axis) to $\pi/4$. For any angle ϕ , ρ ranges from 0 (origin) to $\cos \phi$. The angle θ ranges from 0 to 2π around the z -axis. Thus, $E = \{(\rho, \theta, \phi) \mid 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi/4, 0 \leq \rho \leq \cos \phi\}$. Step 2: Integrate to find volume.

$$\begin{aligned} V(E) &= \iiint_E dV = \int_0^{2\pi} \int_0^{\pi/4} \int_0^{\cos \phi} \rho^2 \sin \phi d\rho d\phi d\theta \\ &= 2\pi \int_0^{\pi/4} \sin \phi \left[\frac{1}{3} \rho^3 \right]_0^{\cos \phi} d\phi = \frac{2\pi}{3} \int_0^{\pi/4} \cos^3 \phi \sin \phi d\phi \end{aligned}$$

Let $u = \cos \phi \implies du = -\sin \phi d\phi$. When $\phi = 0, u = 1$. When $\phi = \pi/4, u = 1/\sqrt{2}$.

$$V(E) = \frac{2\pi}{3} \int_{1/\sqrt{2}}^1 u^3 du = \frac{2\pi}{3} \left[\frac{1}{4} u^4 \right]_{1/\sqrt{2}}^1 = \frac{\pi}{6} \left(1 - \frac{1}{4} \right) = \frac{\pi}{6} \left(\frac{3}{4} \right) = \frac{\pi}{8}$$